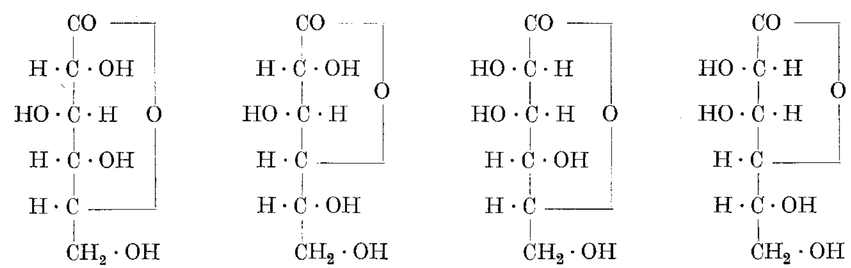
The structure of carbohydrates [1])

by **Walter Norman Haworth.**

(2. IV. 28.)

The selection by *Tollens* and *Fischer* of the five-atom ring structure for glucose and its glucosides was founded upon the analogy of the five-atom ring structure of lactones. Sugars easily pass by bromine oxidation into hexonic acids and there-after by ring closure into lactones, such as γ-gluconolactone and γ-mannonolactone. These lactones are well-defined crystalline products and are easily obtained from the corresponding sugars. *Hudson*[2]) reviewed the properties of no fewer than 24 such lactones, and reached the view that such lactones were constituted as five-atom ring compounds. In 1914 *J. U. Nef*[3]) isolated two new lactones related to glucose and mannose and formulated these as β-lactones having a four-atom ring structure.

It seemed unsatisfactory that the basis of the structural formulation of sugars should remain one of mere analogy, and the work of the Birmingham school of chemistry has in the past few years been directed to the study of lactone structure. It has been shown that the two new lactones isolated by *Nef* are not β-lactones, but should be formulated as δ-gluconolactone and δ-mannonolactone[4]), having the structure of six-atom rings. We therefore have two pairs of lactones derived from glucose and mannose.



δ-Gluconolactone γ-Gluconolactone δ-Mannonolactone γ-Mannonolactone

It was clearly necessary to decide which of these two alternatives represented the ring structure in the parent sugars. The difficulty arises, however, that if the study is confined to unsubstituted lactones, these can pass by isomeric change into each other. When the hydroxyl

[1]) A lecture delivered at the University of Neuchâtel on February 25th, 1928 on the occasion of the half-yearly meeting of the Swiss Chemical Society; published by decision of the Publication Committee.

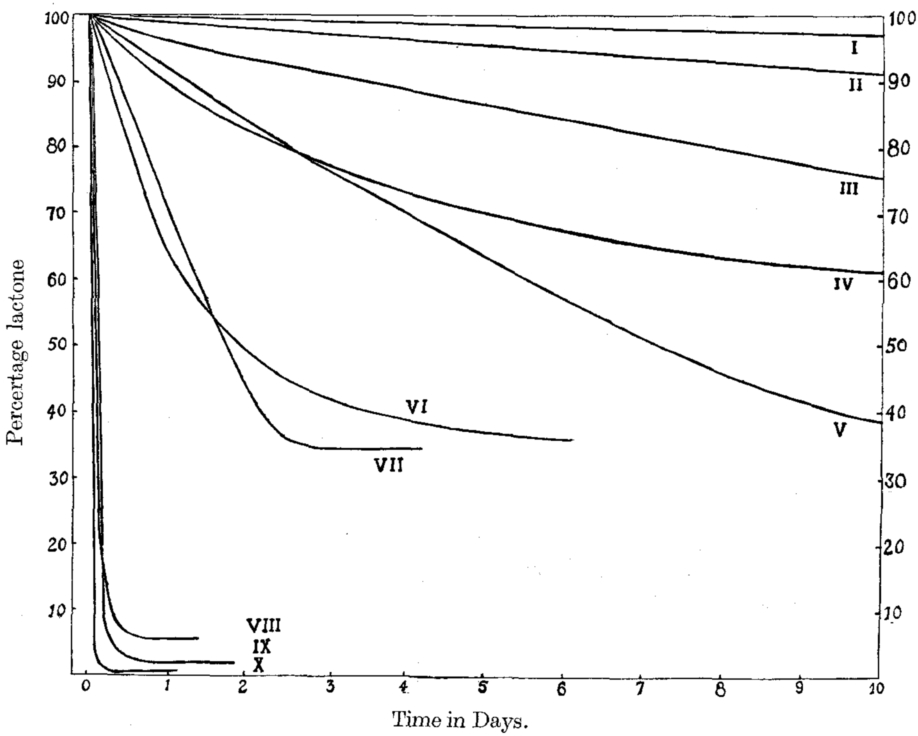
[2]) Am. Soc. **32**, 338 (1910).

[3]) A. **403**, 204 (1914).

[4]) *Haworth* and *Nicholson*, Soc. **1926**, 1899.

groups of a sugar are protected by methyl residues, a series of well defined methylated sugars is made available. Each of these methylated aldoses changes by oxidation with bromine water into the related monobasic acid, and then by desiccation into the lactone. Thus two series of methylated lactones become available, and these have been studied (a) by determining the comparative rates of hydrolysis in water to the related acid, and (b) by direct oxidation and degradation to simpler products.

The curve showing the rate of hydrolysis[1]) of lactones is given below for ten selected cases.

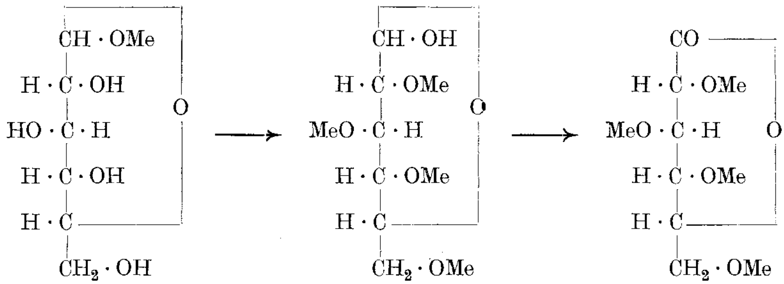


Curves (I) to (V) indicate a slow rate of change and the curves (VI) to (X) a much more rapid change. The former series illustrate the cases of γ-lactones (five-atom rings) and the latter series the δ-lactones (six-atom rings). The individual curves (V) and (X) are the γ- and δ-glucono-lactones and Nos. (I) and (VI) the tetramethyl- γ- and δ-mannono-lactones respectively. Other curves represent the galactono-, xylono-, and arabono-lactones. It was possible to differentiate between the

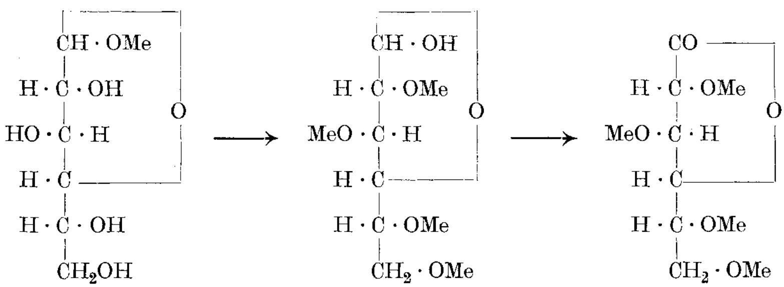
[1]) *Drew, Goodyear* and *Haworth*, Soc. **1927**, 1237; *Haworth* and *Porter*, Soc. **1928**, 611.

two types of rings in lactones and therefore to derive the ring structure[1]) of the methylated sugars which were obtainable from their glucosides.

The ascertained results are illustrated in the case of glucose by the following formulae:



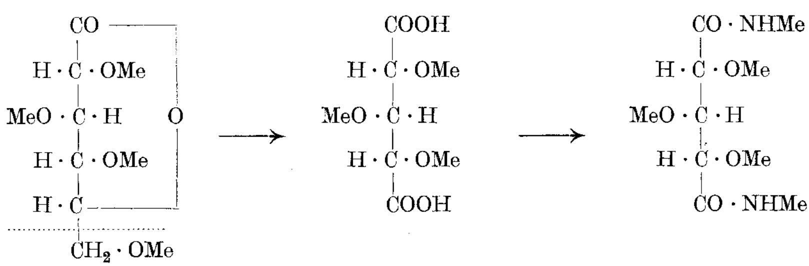
α- and β-Methylglucoside (normal) — Tetramethyl glucose (crystalline, normal) — 2:3:4:6-Tetramethyl δ-gluconolactone



γ-Methylglucoside — Tetramethyl γ-glucose — 2:3:5:6-Tetramethyl γ-gluconolactone (crystalline)

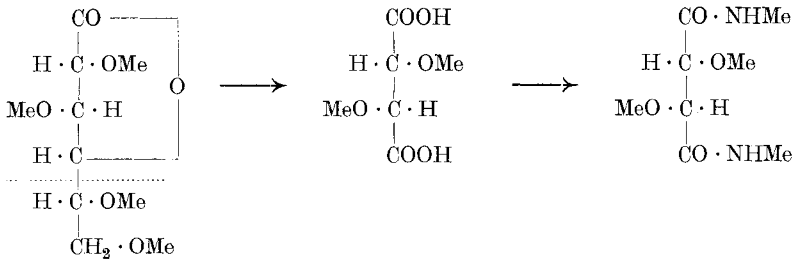
Whilst the δ-lactone is not crystalline it gives a crystalline phenylhydrazide.

Adopting now the method (b), depending on oxidation with nitric acid, each of these lactones was degraded in the following way to simple dibasic acids.



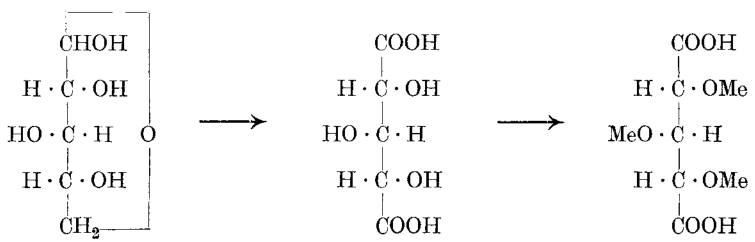
2:3:4:6-Tetramethyl δ-gluconolactone — Xylo-trimethoxyglutaric acid — Crystalline methylamide

[1]) *Haworth, Hirst* and *Miller*, Soc. **1927**, 2436; *Goodyear* and *Haworth*, Soc. **1927**, 3136; *Haworth, Hirst* and *Jones*, Soc. **1927**, 2428.

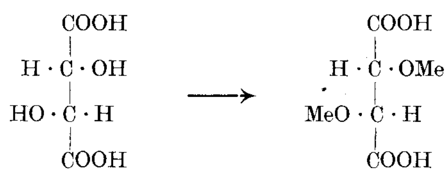


2:3:5:6-Tetramethyl γ-gluconolactone — Dimethyl-d-tartaric acid — Crystalline methylamide

The crystalline methylamides of these acids were compared directly with specimens obtained, (1) by oxidising xylose to trihydroxy-glutaric acid, followed by methylation and conversion to the methylamide; and (2) by methylation of ordinary d-tartaric acid and conversion of this product to the methylamide.



Xylose — Trihydroxy-glutaric acid — Xylo-trimethoxy glutaric acid



d-Tartaric acid — Dimethyl tartaric acid

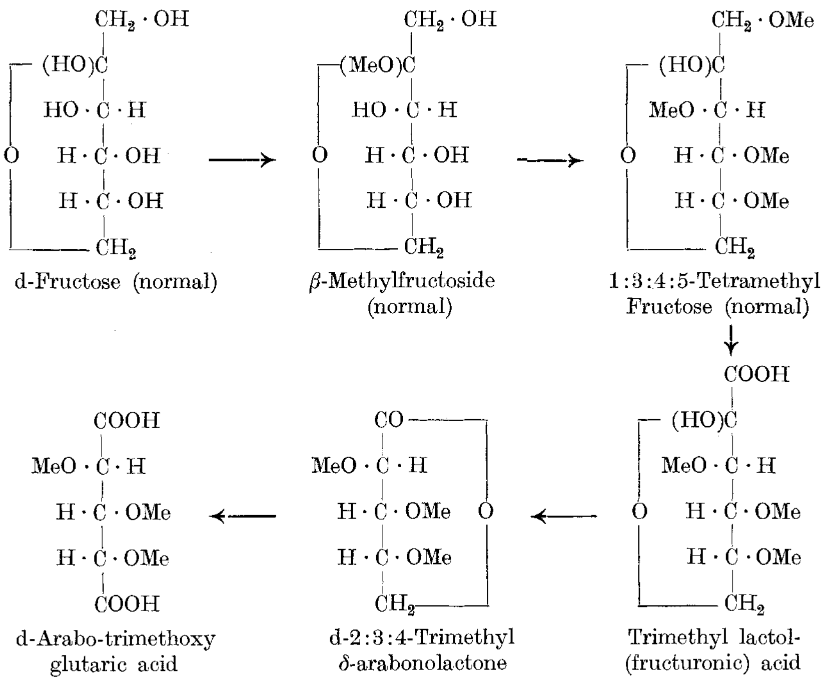
The results furnished by a combination of methods (a) and (b) seem to leave no room for doubt that the crystalline α- and β-methylglucosides (*Fischer*) must now be represented by six-atom ring formulae, and that the γ-methylglucoside discovered by *Fischer* in 1914 has a five-atom ring structure.

These experimental methods have been extended to include all the common aldoses namely, galactose, mannose, xylose, arabinose, and also, by a slight modification of method, identical conclusions have been reached in regard to the structure of the representative ketose, fructose. The crystalline β-methylfructoside which is normally related to crystalline fructose undergoes, on methylation[1]), conversion to a

---

[1]) *Hudson*, Am. Soc. **38,** 1216 (1916); Miss *E. S. Steele*, Soc. **1918**, 257.

tetramethyl methylfructoside, which gives a crystalline tetramethyl fructose. Oxidation of the latter in two stages[1]), first with nitric acid and then oxidation of this product with acid permanganate, leads to a crystalline trimethyl δ-arabonolactone which is converted to a trimethoxy glutaric acid by the agency of nitric acid according to the following scheme:



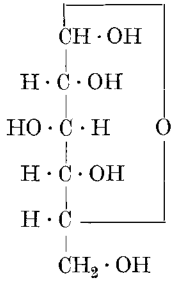
The five-atom ring structure of γ-fructose derivatives will be discussed later when we come to sucrose.

The experimental results point to the conclusion that the structural form of glucose which occurs combined, either in α- and β-methylglucosides or in those disaccharides containing glucose residues, is the six-atom ring type of glucose, the formula for which is represented by (I).

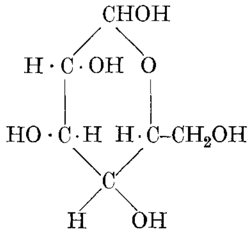
A model constructed on this basis shows that the formula is accurately represented by a hexagon ring, which may be written as formula (II). It will be shown that this hexagon formula introduces many simplifications into the study of sugars. The configuration of the groups is, however, more easily seen by writing this in perspective as in formula (III), where above and below the plane of the six-atom ring are distri-

[1]) *Haworth, Hirst* and *Learner*, Soc. **1927**, 1040.

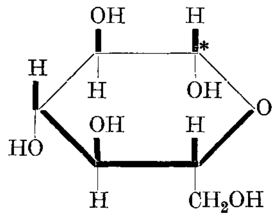
buted the hydrogens and hydroxyls. The reducing group is marked by the asterisk*.



(I)

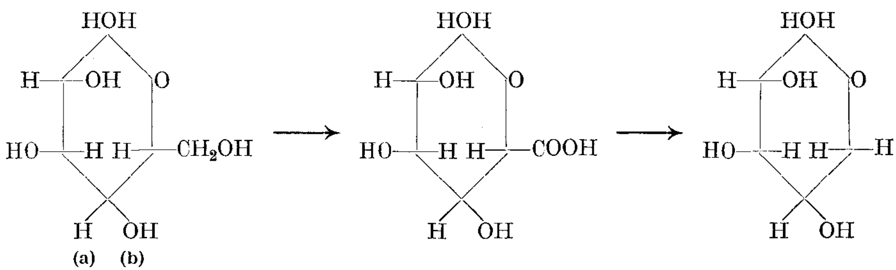


(II)



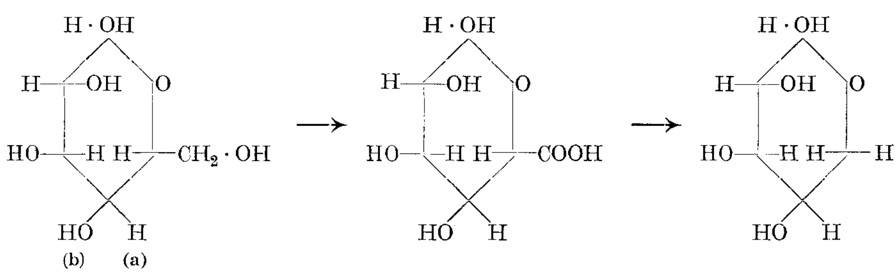
(III) β-Glucose

The relationship of hexoses to pentoses is clearly brought out by considering the reactions of the side chain in glucose or galactose corresponding to the group -CH_2OH.



d-Glucose d-Glycuronic acid Xylose

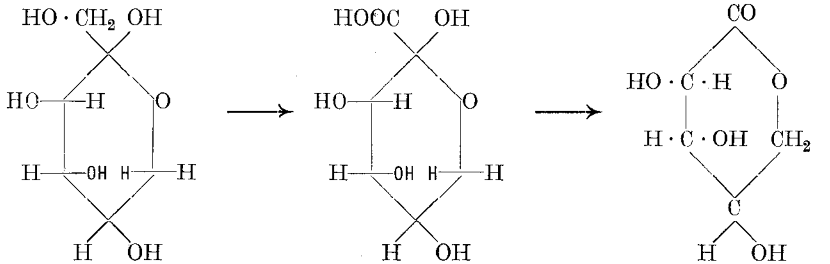
This side-chain appears, as is well-known, in glycuronic acid as a COOH group, illustrating the analogy of glucose and glycuronic acid with, for example, benzyl alcohol and benzoic acid. Just as the latter can pass easily into benzene, so glycuronic acid appears to undergo loss of carbon dioxide to give xylose. This will explain the occurrence of d-xylose in nature and particularly the existence of xylose chains in the xylan portion of esparto cellulose. Similarly the occurrence in fruit gums of both d-galactose and l-arabinose is illustrated by the following conversion.



d-Galactose d-Galacturonic acid l-Arabinose

It may be observed that glucose is probably the first sugar produced by nature in the plant, and its conversion to galactose involves the inversion of the groups marked (a) and (b) above.

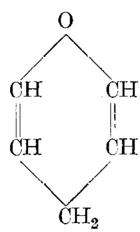
Similarly the ketose, fructose, is represented by a hexagon formula having, however, a side-chain $-CH_2OH$ attached at a different position, namely at the reducing group in the molecule.



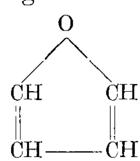
d-Fructose d-Fructuronic acid d-Arabono-δ-lactone

Here again, it is the side-chain which even in the methylated sugar is the first to suffer attack by conversion to a carboxyl group, and elimination of this acid residue leads to the formation of the related d-arabonolactone.

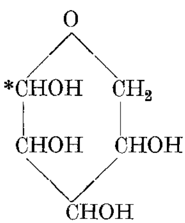
The generalisation is therefore reached that the five- and six-atom ring forms of sugars are related to the parent compounds pyran and furan, which in their reduced and hydroxylated forms appear in formulae (A) and (B) below, and which by a shortened nomenclature may be named Pyranose and Furanose, the latter being the simplest possible tetrose.



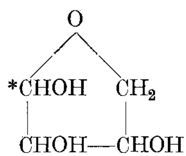
(Pyran)



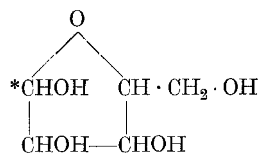
(Furan)



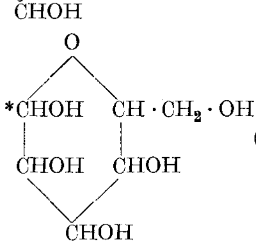
(A)



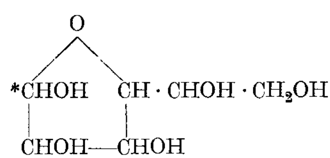
(B)



(C)



(D)

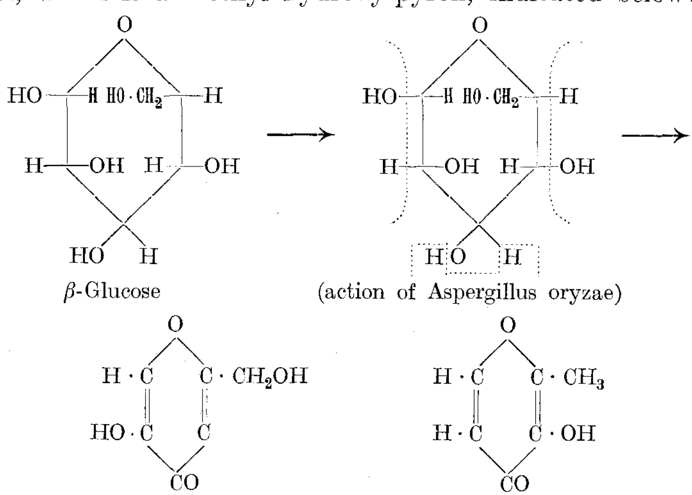


(E)

The pyranoses will then be ordinary sugar forms related to α- and β-methylglucoside, β-methylxyloside, α- and β-methylarabinoside, and α- and β-methylgalactoside. The furanose forms, however, are the labile or so-called γ-sugars. The attachment of a side-chain to the pentose (A) illustrates clearly the structure of a typical hexose (D) (glucose, galactose, etc.), whilst a side-chain introduced into (B) shows the γ-series of the pentoses (C) and a lengthened side-chain the corresponding labile or γ-hexoses (E). If this nomenclature be adopted it will be possible to dispense with much of the confusing terminology associated with the sugar group. Both the structure and stereochemistry will be clearly defined by the expressions gluco-pyranose, xylo-pyranose, etc. for the typical hexose and pentose, glucose and xylose, whilst the labile or γ-forms of these will be recognisable as gluco-furanose, xylo-furanose, etc.

The ease with which certain sugars pass into furfuraldehyde by the agency of mineral acids has sufficed to relate some of the sugars to furan. In this connection it may be added that tetramethyl γ-fructose changes on attempted acetylation or with dilute mineral acids to a derivative of furfuraldehyde. On the other hand, the close relationship of glucose as a six-atom ring structure to substances of the nature of pyran is seen in the ready transformation of glucose by the action of Aspergillus oryzae to a derivative of pyron, namely kojic acid, which appears when the latter organism is grown on koji or steamed rice. In contact with glucose this organism converts one-tenth of the sugar into kojic acid[1]).

Again, by heating malt one of the products which can be isolated is maltol, which is a methyl hydroxy-pyron, indicated below:

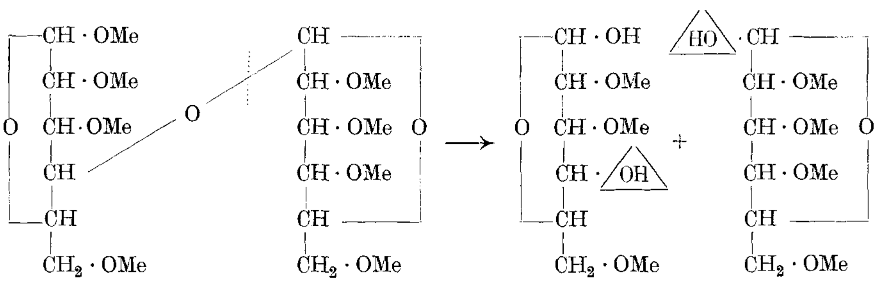


β-Glucose (action of Aspergillus oryzae) Kojic acid Maltol

[1]) *Yabuta,* J. Chem. Soc. Tokyo **37,** 1185, 1234 (1916).

Disaccharides.

The determination of the structure of disaccharides has proceeded on the following plan. The eight available hydroxyl groups in the formula $C_{12}H_{22}O_{11}$ may be protected by methylation methods, using methyl sulphate and alkali as reagents and completing the methylation by *Purdie's* reagents (methyl iodide and silver oxide). Hydrolytic cleavage of the fully substituted disaccharide then yields two components which can be separated. The determination of the structure of each of these components leads to a preliminary decision as to the structure of the biose. Thus, octamethyl maltose, octamethyl cellobiose and octamethyl lactose may be formulated as below, hydrolysis of each of these yielding the products shown.



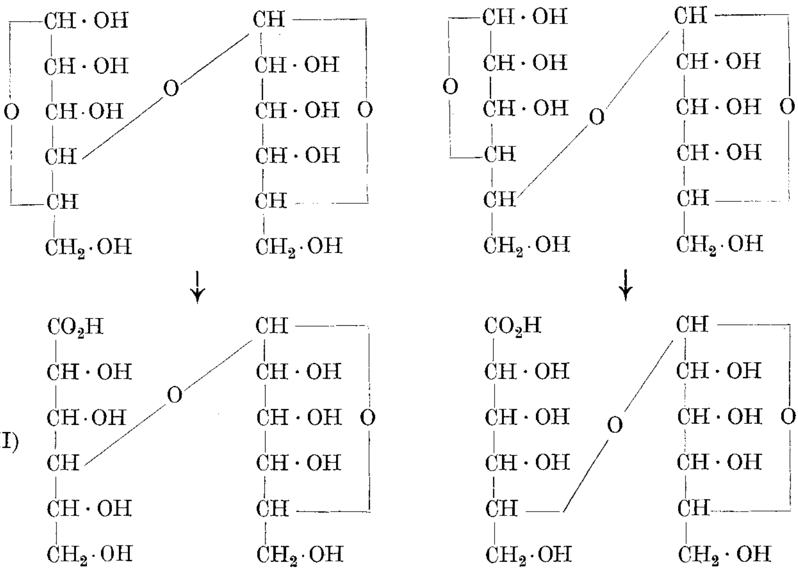
Octamethyl maltose
,, cellobiose
,, lactose

2:3:6-Trimethyl Glucose

2:3:4:6-Tetramethyl glucose (from maltose and cellobiose)
2:3:4:6-Tetramethyl galactose (from lactose)

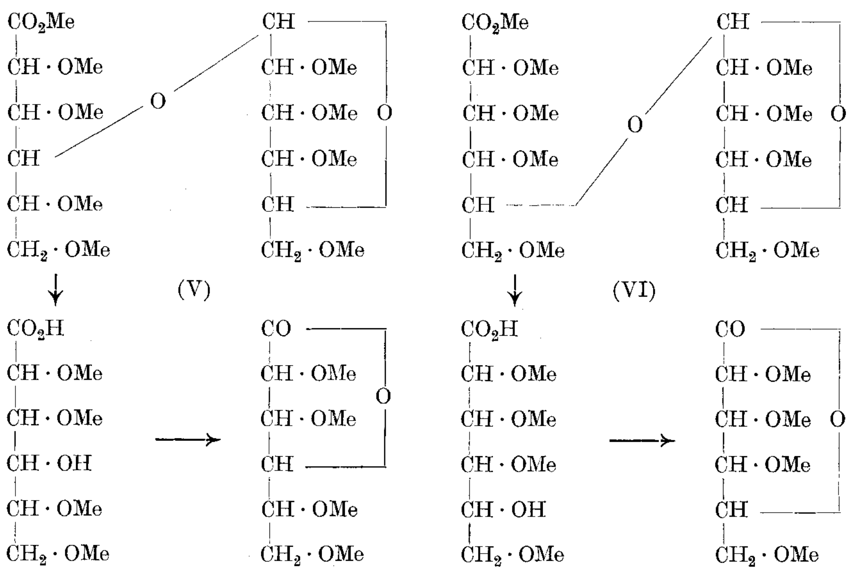
The exposed hydroxyl groups in the two cleavage fragments indicate the probable position of the biose linking. Whilst the tetramethyl sugars (glucose or galactose) can definitely be shown to have this structure by the work previously outlined, the constitution of the trimethyl glucose fragment leads to some dubiety since the groupings at carbon atoms 4 and 5 in the chain might be reversed, so that the free hydroxyl group is at position 5 and the oxide-ring linking at position 4. In such a case it would be conceivable that one of the hexose components in each of these disaccharides might be a furanose or five-atom ring sugar. That this is not the case, however, and that the components in each fragment are six-atom ring sugars may be seen from the succeeding experiments.

The alternative structures for these three disaccharides indicated by the above preliminary results may be represented by formulae (I) and (II).



(I) (II) (III) (IV)

Now each of the disaccharides maltose, cellobiose and lactose has been oxidised to give the corresponding bionic acid (maltobionic, cellobionic, lactobionic), which again may be represented either by formula (III) or (IV). Methylation of these bionic acids would give products in each case which may be represented by the alternative formulae (V) and (VI).

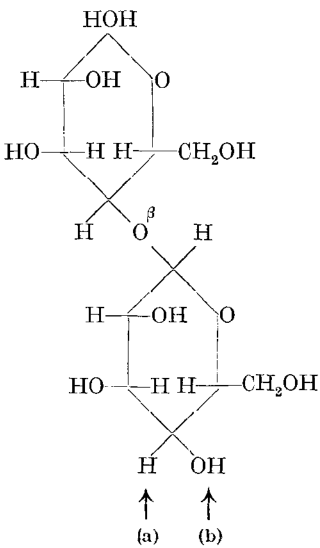


(V) (VI)

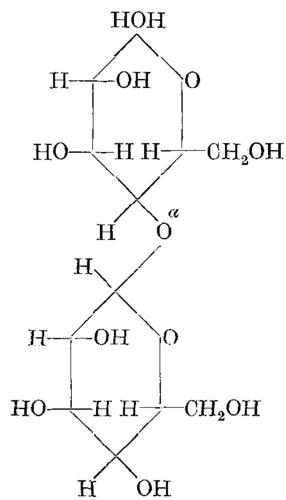
(VII) (IX) (VIII) (X)

Hydrolysis of either (V) or (VI) would lead to the acid (VII) or (VIII) and finally to the lactone (IX) or (X). Thus the choice between the disaccharide formulae (I) and (II) would be determined by the diagnosis of the lactone (IX) as a γ-lactone or the lactone (X) as a δ-lactone.

The alternatives presented in this way have been fully tested by methylation of the three bionic acids[1]), and in each case the lactone isolated was 2 : 3 : 5 : 6-tetramethyl γ-gluconolactone, a crystalline substance indicated by formula (IX). The constitution of this lactone has been discussed in the earlier paragraphs. It follows therefore that maltose, cellobiose and lactose are to be represented by formula (I) and not by formula (II). It is possible then to write the constitution of these sugars in the following way:



Cellobiose
In Lactose, groups (a) and (b) of cellobiose are transposed



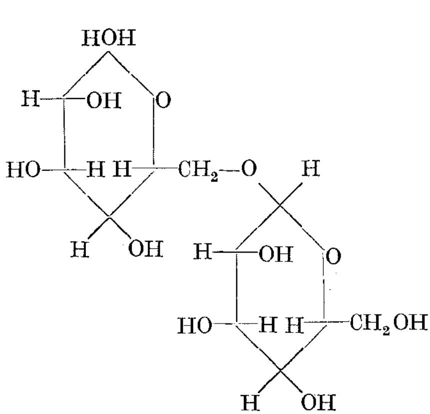
Maltose

By methods which are analogous to those described already, the constitutional formulae of gentiobiose and melibiose have also been determined[2]). These sugars are represented by formulae below, wherein it is seen that the non-reducing hexose residue is attached to a hydroxyl group in the side-chain of the first hexose residue.

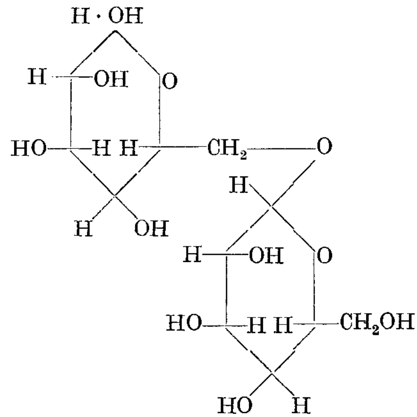
---

[1]) *Haworth* and *Peat*, Soc. **1926,** 3094; *Haworth* and *Long*, Soc. **1927,** 544; *Haworth, Long* and *Plant*, Soc. **1927,** 2809.

[2]) *Haworth* and *Wylam*, Soc. **1923,** 3120; *Haworth, Loach* and *Long*, Soc. **1927,** 3146.

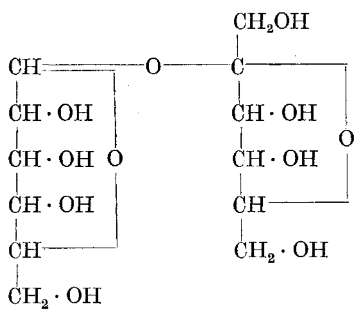


Gentiobiose



Melibiose

The problem of the constitution of sucrose has presented many difficulties, both theoretical and experimental, and the structural formula assigned to the most important of the disaccharides is indicated by:



(Sucrose)

As is wellknown sucrose, having $[\alpha]_D$ $+66.5^0$, gives on hydrolysis invert sugar having $[\alpha]_D$ -20.5^0. On the other hand, it was noticed[1]) that octamethyl sucrose, $[\alpha]_D$ $+66.5^0$, gave a mixture of methylated glucose and fructose, having $[\alpha]_D$ $+56.5^0$, so that by hydrolysis no inversion of sign occurs. Working with heptamethyl sucrose one was able to isolate[2]) the fructose component as a tetramethyl fructose having $[\alpha]_D$ $+31^0$, whilst from octamethyl sucrose on hydrolysis the crystalline tetramethyl glucose having $[\alpha]_D$ $+82^0$ was also isolated. This fructose component was shown to be a derivative of labile or γ-fructose, and thus we can represent the changes in structure which occur in the production of fructose from sucrose in the following way:

[1]) *Haworth* and *Law*, Soc. **1916**, 1314.
[2]) *Haworth*, Soc. **1920**, 199.

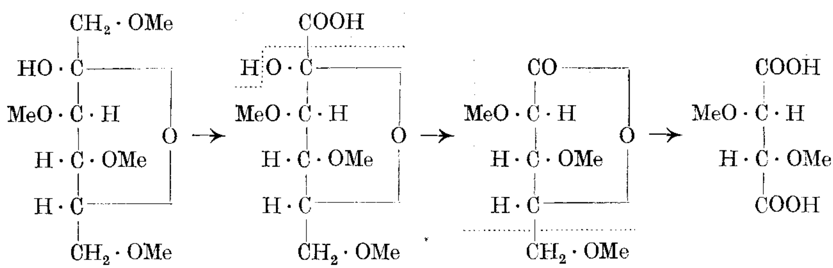
Sucrose $\longrightarrow$ glucose (normal) + fructose (labile)

$\downarrow$

crystalline fructose (normal)
on isolation

The difficulties involved in isolating a pure specimen of tetramethyl γ-fructose from methylated sucrose were successfully overcome, and this was submitted to careful examination with the view of determining its structure. The results obtained are set out below.

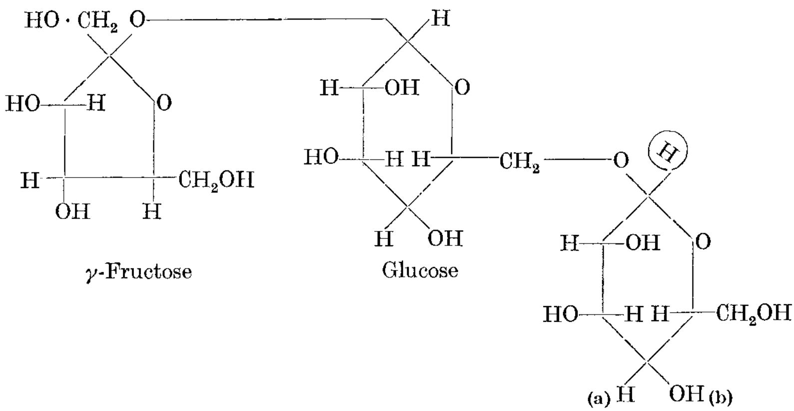
Oxidation of the tetramethyl γ-fructose gave a trimethyl lactol acid, showing that one of its methylated side-chains had undergone conversion to a carboxyl group[1]. Crystalline derivatives of this lactol acid were compared. It was then further oxidised by means of acid permanganate and gave in good yield the 2 : 3 : 5-trimethyl γ-arabonolactone which was crystalline. The structure of this lactone was ascertained by methods which have already been outlined, and it was shown that it was degraded by nitric acid oxidation to give l-dimethyl tartaric acid, which in the form of its crystalline methylamide was identical with a specimen prepared for this comparison from l-tartaric acid.



1:3:4:6-Tetramethyl γ-fructose — Trimethyl lactol acid — d-2:3:5-Trimethyl γ-arabonolactone — l-Dimethyl tartaric acid

It is thus seen that sucrose contains a gluco-pyranose residue linked with a fructo-furanose residue through the reducing positions. This conclusion may now be combined with our previous ascertained knowledge of the structure of gentianose and raffinose, since these latter trisaccharides may be made to yield either sucrose on the one hand by enzyme hydrolysis, or gentiobiose respectively by a different enzyme. Since the structure of the three related bioses has now been determined, it is possible to represent the two trisaccharides by the formula indicated below.

[1] *Haworth* and *Hirst*, Soc. **1926**, 1858; *Haworth, Hirst* and *Nicholson*, Soc. **1927**, 1513; *Avery, Haworth* and *Hirst*, Soc. **1927**, 2308; *Haworth, Hirst* and *Learner*, Soc. **1927**, 2432.



γ-Fructose

Glucose

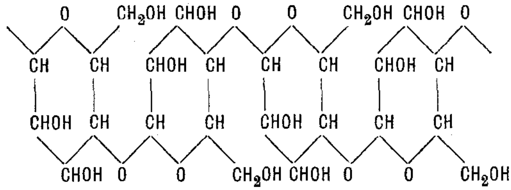
Glucose [in Gentianose]
Galactose (a, b and (H) inverted)
[in Raffinose]

Future work is being directed to the problem of the structure of polysaccharides. In this field there are many speculations and numerous rival formulae; but it may be claimed that the determination of structure in the disaccharide series lends valuable aid to this problem. The more recent results show that cellulose cannot be considered to be a polymerised form of (1 : 4) (1 : 5) anhydro-glucose, since the trimethyl derivative of this anhydro-glucose has been prepared by *Hess*[1]) and it appears to have no relationship to trimethyl cellulose. The X-ray data furnished by the work of *O. L. Sponsler* are of considerable interest in that they tend to confirm the presence of a six-atom ring (glucopyranose) in cellulose. *Sponsler* and *Dore*[2]) have interpreted these data on the plan of continuous chains of glucose units alternately linked through the groupings 1 : 1 and 4 : 4. This would imply that the type of linking which occurs in the disaccharide cellobiose is not pre-formed in cellulose. It seems unnecessary, however, to disregard at this stage of our knowledge the relevant fact that over 50% of cellobiose can be formed from cellulose by acetolysis. Reconsideration of the X-ray data leads one to suggest that they accommodate the cellobiose type of structure at least equally as well as the linking which is preferred by the former authors. In this connection the *trans Sachse* model for a six-atom ring may be tentatively adopted for the β-glucose unit in the crystalline polymeride. It is then possible to build up models on the basis of the cellobiose structure (1 : 4, 1 : 4), and representing a regular pattern which may be indicated as follows:

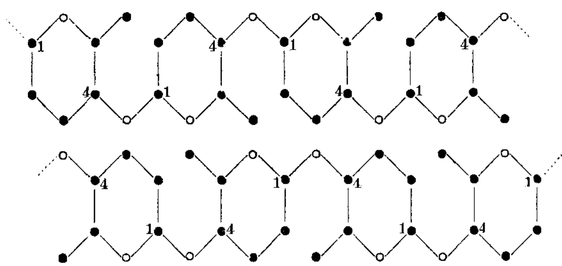
---

[1]) B. **60,** 1898 (1927); compare *Freudenberg,* A. **460,** 295 (1928)).

[2]) Colloid Symposium Monograph (New York) **1926,** No. 4, 174; Compare *Nature,* **120,** 767 (1927).



It will be seen that the intervening spaces between each hexose unit also approximate to a hexagonal figure. These in the solid state would hardly be distinguishable from the outlines of the hexose units themselves. In alternate chains these imperfect hexagonal spaces may be regarded as being covered by the hexagon unit of glucose. The whole presents a uniform pattern which may be represented in outline as follows:



Birmingham, Department of Chemistry, The University.

Zur Pufferung der Fäces. Pufferstudien X[1])

von **M. Kartagener.**

(4. IV. 28.)

Die Wasserstoffionen-konzentration der menschlichen Fäces war schon mehrfach Gegenstand klinischer Untersuchungen, wenn auch nicht so häufig wie die von Blut, Urin und andern Körperflüssigkeiten. Die Befunde, die die verschiedenen Untersucher dabei erhoben haben, stimmen im grossen Ganzen überein, obschon ihre Deutung je nach der Fragestellung, von der die Untersuchung ausgeht, zu ganz verschiedenen Ergebnissen führte. Während von pädiatrischer Seite[2]) aus der $H^{\cdot}$-ionenkonzentration der Fäces Rückschlüsse auf die bakteriellen Vorgänge der Gärung und Fäulnis, die sich im Darmlumen abspielen, gezogen

[1]) Pufferstudien IX. *K. Klinke* und *F. Leuthardt*, Klin. Wochenschr. **6**, 2409 (1927).
[2]) Vgl. *Freudenberg* und *Heller*, Jahrb. Kinderheilk. **94**, 251 (1921); *Scheer* und *Müller*, Jahrb. Kinderheilk. **101**, 143 (1923) und **102**, 93 (1924).